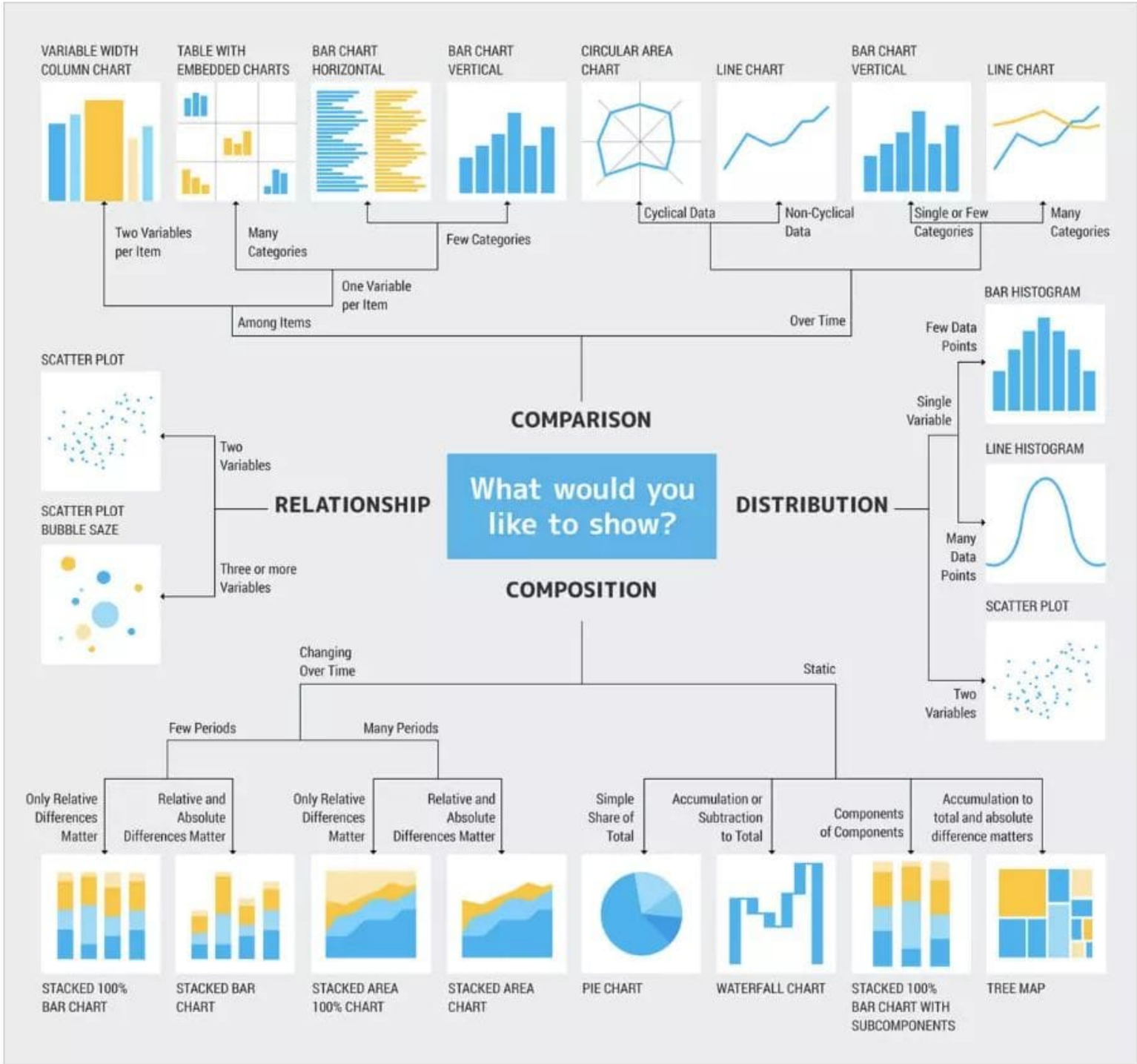
Data visualisation

The Three Jobs of a Visualization

Explore — find what you didn't know to look for

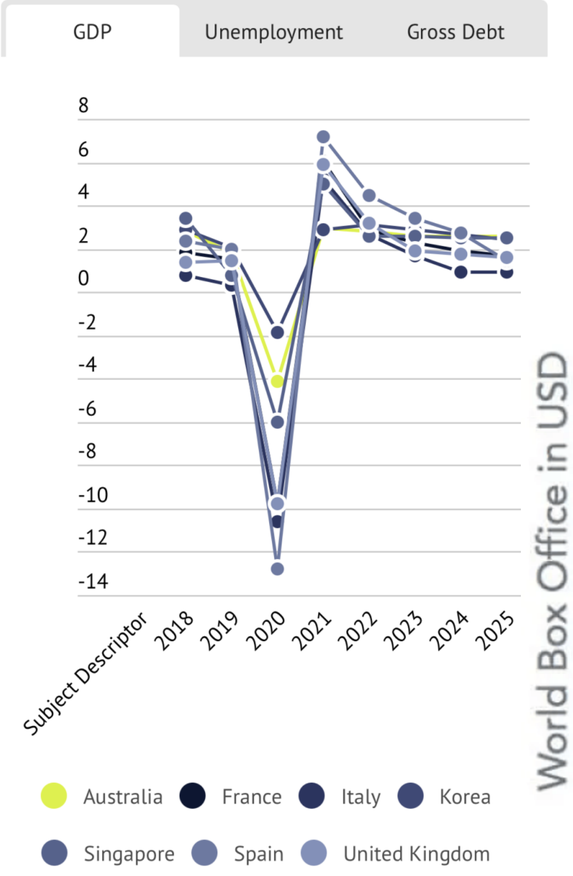
Explain — make a known finding unmissable

Evaluate — let a stakeholder act on it

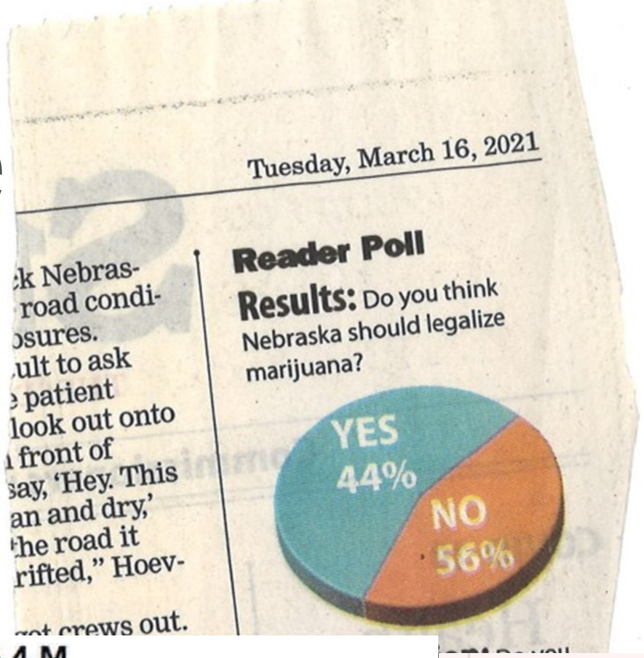

What would you like to show?
COMPARISON
VARIABLE WIDTH COLUMN CHART
TABLE WITH EMBEDDED CHARTS
BAR CHART HORIZONTAL
BAR CHART VERTICAL
CIRCULAR AREA CHART
LINE CHART
BAR CHART VERTICAL
LINE CHART
Two Variables per Item
Many Categories
Few Categories
One Variable per Item
Among Items
Cyclical Data
Non-Cyclical Data
Single or Few Categories
Many Categories
Over Time
RELATIONSHIP
SCATTER PLOT
SCATTER PLOT BUBBLE SAZE
Two Variables
Three or more Variables
DISTRIBUTION
BAR HISTOGRAM
Few Data Points
Single Variable
LINE HISTOGRAM
Many Data Points
SCATTER PLOT
Two Variables
COMPOSITION
Changing Over Time
Static
Few Periods
Many Periods
Only Relative Differences Matter
Relative and Absolute Differences Matter
Only Relative Differences Matter
Relative and Absolute Differences Matter
Simple Share of Total
Accumulation or Subtraction to Total
Components of Components
Accumulation to total and absolute difference matters
STACKED 100% BAR CHART
STACKED BAR CHART
STACKED AREA 100% CHART
STACKED AREA CHART
PIE CHART
WATERFALL CHART
STACKED 100% BAR CHART WITH SUBCOMPONENTS
TREE MAP

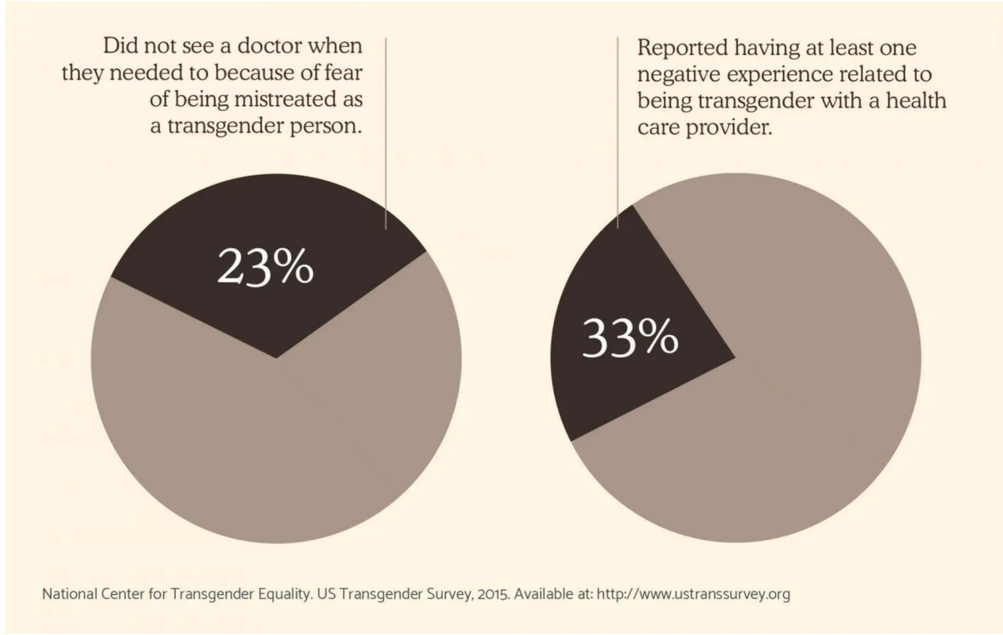
Source

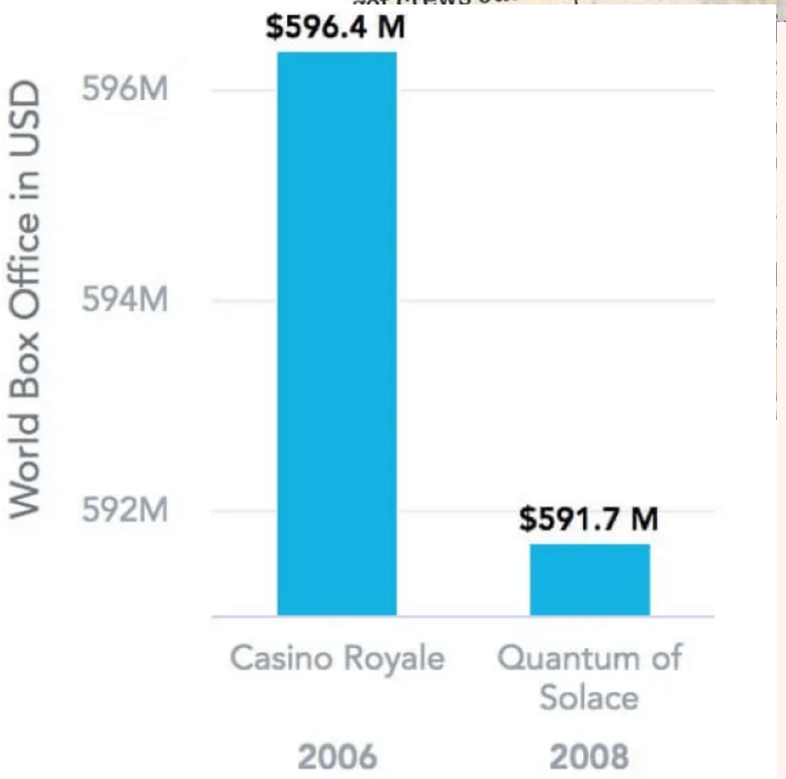
List of shame

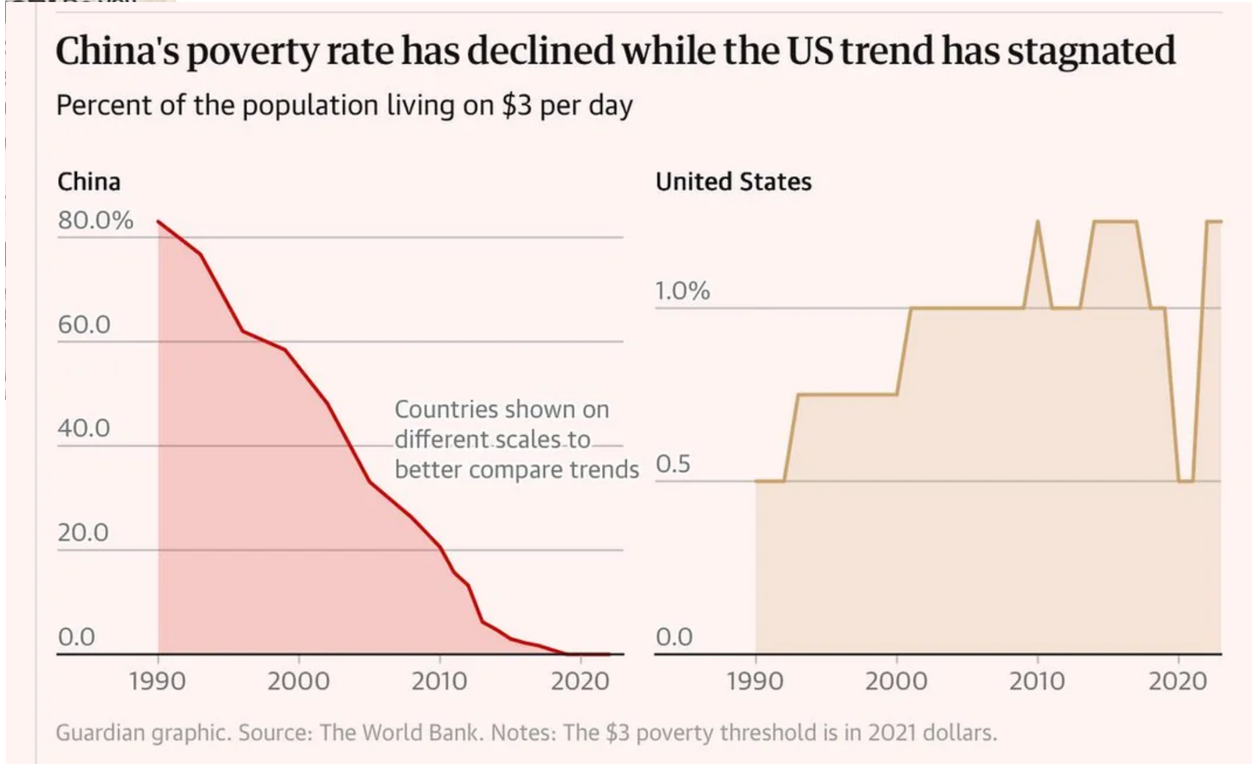


Source: International Monetary Fund, World Economi Outlook Database, October 2020









Important to remember

1. Create visualizations with a specific target audience in mind
2. Use your colors carefully - up to 6-8 colors per visualization, and use them to reinforce a message, not tell a story
3. Avoid excessive detail and cluttering up your visualizations
4. Highlight only the most important information in the visualization
5. Use legends and tooltips to explain what is happening in the visualization
6. Provide context for your data whenever possible
7. Structure the visualization so that the most important information is displayed first

Checklist

- Does the chart survive being printed in greyscale?
- Does it answer a question someone asked?
- Is uncertainty visible, not just computed?
- Would a tired judge understand it?